Sequences and recurrences

L4-X8 Level 4 Olympiad · Algebra · Beyond the textbook

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Linear recurrences and the characteristic equation, integer terms from nonlinear recurrences, periodicity, sharp bounds, strong induction, and recurrences hidden in counting.
By the end you can solve linear recurrences, including repeated and complex roots, and use the roots to prove properties of the terms; prove that a nonlinear recurrence gives integers by finding a hidden linear recurrence; use periodicity modulo m, power comparisons and counting models to answer questions about sequences.
Prerequisites: Functional equations (L4-AL1), Polynomials and Vieta's formulas (L3-AL1). **Contests:** BMO1, BMO2, MOG, AIME, USAJMO. **Time:** about 120 minutes.

1 Lesson

Prove that $\lfloor (2+\sqrt{3})^n \rfloor$ *is odd for every* $n \geq 1$. Put $a_n = (2+\sqrt{3})^n + (2-\sqrt{3})^n$. Both $2 \pm \sqrt{3}$ are roots of $x^2 - 4x + 1$, so $a_{n+2} = 4a_{n+1} - a_n$ with $a_0 = 2$ and $a_1 = 4$. Hence every a_n is an even integer. Since $0 < (2-\sqrt{3})^n < 1$, we get $\lfloor (2+\sqrt{3})^n \rfloor = a_n - 1$, which is odd.

De Moivre solved linear recurrences in 1730, and Euler and Lagrange gave the characteristic equation used below. School courses stop at arithmetic and geometric sequences. Olympiad problems expect three further habits, and this unit is built around them. Replace a nonlinear recurrence by a linear one, read a recurrence modulo m, and compare a sequence with a power of itself to get sharp bounds. You will use all three in the practice problems.

1.1 Linear recurrences

Theorem 1.
Let $c_k \neq 0$ and $p(x) = x^k - c_1 x^{k-1} - \cdots - c_k$. The sequences with

$$a_{n+k} = c_1 a_{n+k-1} + \cdots + c_k a_n \quad (n \geq 0)$$

form a vector space of dimension k, and each is fixed by $a_0, \ldots, a_{k-1}$. If r is a root of p of multiplicity m, the sequences $n^j r^n$ with $0 \leq j < m$ are solutions, and these k sequences, taken over all roots, form a basis. In particular, if p has distinct roots $r_1, \ldots, r_k$, then $a_n = A_1 r_1^n + \cdots + A_k r_k^n$.

Proof. Let E be the shift, $(Ea)_n = a_{n+1}$, so the recurrence reads $p(E)a = 0$. The map $a \mapsto (a_0, \ldots, a_{k-1})$ is a bijection, which gives the dimension. Write $p(x) = q(x)(x-r)^m$. Then

$$(E-r)\big(n^j r^n\big) = r^{n+1}\big((n+1)^j - n^j\big),$$

which is r^n times a polynomial of degree $j-1$, so $(E-r)^m$ kills $n^j r^n$ for $j < m$.

For independence, suppose $\sum_i P_i(n) r_i^n = 0$ for all n, with distinct roots r_i (all nonzero as $c_k \neq 0$) and polynomials P_i not all zero, and choose such a relation with the least total degree. A single

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term $P(n)r^n = 0$ forces $P = 0$. Otherwise apply $E - r_1$. The term with r_1 becomes $r_1^n \cdot r_1\big(P_1(n+1) - P_1(n)\big)$, of smaller degree, while for $i \neq 1$ the term $P_i(n)r_i^n$ becomes $Q_i(n)r_i^n$ with Q_i of the same degree as P_i and leading coefficient multiplied by $r_i - r_1 \neq 0$. This is a smaller nontrivial relation, a contradiction. ■

Only the independence argument has real content. If you are happy to take the basis on trust, read on and come back to the proof later. What you must know is the statement, and in particular the factor n^j for a repeated root.

For real sequences, a pair of complex roots $\rho e^{\pm i\theta}$ gives the real solutions $\rho^n \cos n\theta$ and $\rho^n \sin n\theta$.

1.2 Integer terms from nonlinear recurrences

Lemma 2 (linearisation).
Let $a_0, a_1, \ldots$ be nonzero with $a_{n+1}a_{n-1} - a_n^2 = C$ for all $n \geq 1$, where C is a constant. Then $K = \frac{a_{n+1}+a_{n-1}}{a_n}$ does not depend on n, so $a_{n+1} = Ka_n - a_{n-1}$.

Proof. Subtracting the relations for n and $n+1$ gives $a_{n+2}a_n - a_{n+1}^2 = a_{n+1}a_{n-1} - a_n^2$, that is $a_n(a_{n+2} + a_n) = a_{n+1}(a_{n+1} + a_{n-1})$. Divide by $a_n a_{n+1}$. ■

If K and the first two terms are integers, every term is an integer. The lemma is short, but spotting the constant C in a problem is the hard part, so always compute $a_{n+1}a_{n-1} - a_n^2$ for the first few terms. The same subtraction works when the relation has extra terms, as in Problem 1.

1.3 Periodicity

Theorem 3.
(1) If a_{n+k} is determined modulo m by $a_n, \ldots, a_{n+k-1}$ modulo m, then $(a_n \bmod m)$ is eventually periodic. If also a_n is determined modulo m by $a_{n+1}, \ldots, a_{n+k}$, the sequence is periodic from the start. (2) If the characteristic polynomial p divides $x^N - c$, then $a_{n+N} = c\,a_n$ for all n.

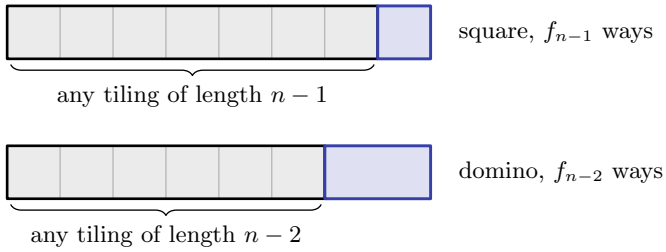
Proof. (1) There are m^k possible windows $(a_n, \ldots, a_{n+k-1}) \bmod m$, so two windows agree, and from then on the sequence repeats. Running the recurrence backwards carries the repetition back to $n = 0$. (2) If $x^N - c = p(x)q(x)$, then $(E^N - c)a = q(E)p(E)a = 0$. ■

For instance $a_{n+2} = -a_{n+1} - a_n$ has $p(x) = x^2 + x + 1$, a factor of $x^3 - 1$, so every solution has period 3. A linear recurrence with integer coefficients and $c_k = \pm 1$ can always be run backwards modulo m, so part (1) gives a period from the start.

1.4 Recurrences in counting

To count objects of size n, condition on the last piece. Tilings of a $1 \times n$ strip by squares and dominoes end in a square or a domino, so their number f_n satisfies $f_n = f_{n-1} + f_{n-2}$, and f_n is the Fibonacci number F_{n+1}.

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When removing the last piece leaves a shape of a different kind, give that shape its own sequence and write a system. For tilings of a $2 \times n$ board by dominoes and L-trominoes, the second sequence counts boards with one corner cell missing. Once the recurrence is known, Theorem 1 gives a closed form and Theorem 3 gives residues.

The other direction is just as useful. A recurrence with positive integer coefficients often counts something, and a count can be read in a second way, by symmetry or by rotation, as in Problem 6.

1.5 Bounds and strong induction

This is the least mechanical part of the unit, since the right comparison has to be guessed. For $a_{n+1} = a_n + f(a_n)$ with f small, look for a power k such that $a_{n+1}^k - a_n^k$ is almost constant, then add up, as in Example 3. A bounded monotone sequence converges, which we quote from analysis. When a_n depends on many earlier terms, assume the claim for every smaller index (strong induction).

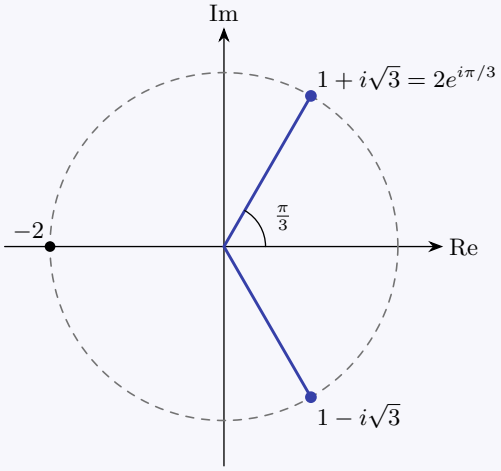
1.6 How to spot the method

1. A linear recurrence with constant coefficients. Write down $p(x)$ and its roots.
2. A quotient such as $\frac{a_n^2+5}{a_{n-1}}$ that should be an integer. Look for a constant combination, then use Lemma 2.
3. A question about residues, or a recurrence that looks trigonometric. Look for a period.
4. An estimate of a_{1000}. Compare a_{n+1}^k with a_n^k.
5. A counting problem with a size n. Condition on the last piece to get a recurrence.

1.7 Worked examples

Example 1.
Let $a_0 = 0$, $a_1 = 1$ and $a_{n+2} = 2a_{n+1} - 4a_n$. Prove that $a_n = 0$ if $3 \mid n$ and $|a_n| = 2^{n-1}$ otherwise.

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Solution.
The terms begin $0, 1, 2, 0, -8, -16, 0, 64$, which suggests $a_{n+3} = -8a_n$. To prove the guess, apply the recurrence twice:

$$a_{n+3} = 2a_{n+2} - 4a_{n+1} = 2(2a_{n+1} - 4a_n) - 4a_{n+1} = -8a_n.$$

So $a_{3j} = 0$, $a_{3j+1} = (-8)^j$ and $a_{3j+2} = 2(-8)^j$, which gives $|a_n| = 2^{n-1}$ when $3 \nmid n$.
Theorem 3(2) explains the guess. The roots of $x^2 - 2x + 4$ are $1 \pm i\sqrt{3} = 2e^{\pm i\pi/3}$, so $x^2 - 2x + 4$ divides $x^3 + 8$. Theorem 1 gives the formula $a_n = \frac{2^n}{\sqrt{3}} \sin \frac{n\pi}{3}$ as well.
Remark. The cube of each root is -8, and that is where the period 3 and the factor -8 come from.

Example 2.
Let $a_0 = 1$, $a_1 = 2$ and $a_{n+1} = \frac{a_n^2+5}{a_{n-1}}$ for $n \geq 1$. Prove that every term is an integer and that consecutive terms are coprime.

Solution.
A direct induction would need $a_{n-1} \mid a_n^2 + 5$, which says nothing about how a_{n-1} and a_n are related. Instead, every term is positive, since numerator and denominator are, and the definition says $a_{n+1}a_{n-1} - a_n^2 = 5$. So Lemma 2 applies, and $a_2 = 9$ gives $K = \frac{9+1}{2} = 5$, that is

$$a_{n+1} = 5a_n - a_{n-1}.$$

Hence all terms are integers.
A common divisor of a_n and a_{n+1} divides $a_{n+1}a_{n-1} - a_n^2 = 5$. Modulo 5 the recurrence reads $a_{n+1} \equiv -a_{n-1}$, so the residues run $1, 2, 4, 3, 1, 2, \ldots$ and are never 0. Hence consecutive terms are coprime. ■
Remark. The constant 5 was never used beyond being constant. Any recurrence of this shape is linear in disguise.

Example 3.
Let $a_1 = 1$ and $a_{n+1} = a_n + \frac{2}{a_n^2}$. Find $\lfloor a_{1000} \rfloor$.

Solution.

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Adding up the increments $\frac{2}{a_n^2}$ directly goes nowhere, since each increment depends on the term we are trying to estimate. Since $(a^3)' = 3a^2$, cube instead:

$$a_{n+1}^3 = a_n^3 + 6 + \frac{12}{a_n^3} + \frac{8}{a_n^6}.$$

Hence $a_n^3 \geq 1 + 6(n-1) = 6n - 5$, and $a_{1000}^3 \geq 5995 > 18^3 = 5832$.
For the upper bound, $a_k^3 \geq 6k - 5 \geq k$ gives $\frac{12}{a_k^3} + \frac{8}{a_k^6} \leq \frac{20}{k}$, so with $H_N = 1 + \frac{1}{2} + \cdots + \frac{1}{N} \leq 1 + \ln N$,

$$a_{1000}^3 \leq 5995 + 20H_{999} \leq 5995 + 20(1 + \ln 999) < 5995 + 159 < 19^3 = 6859.$$

So $\lfloor a_{1000} \rfloor = 18$.
Remark. The power 3 is chosen so that the main term of $a_{n+1}^3 - a_n^3$ is constant. The error terms are then summable with room to spare.

Example 4.
A staircase of n steps is climbed by steps of 1 or 2, never two 2s in a row, and c_n is the number of ways ($c_0 = 1$). Prove that every positive integer is a sum of distinct numbers c_k, $k \geq 1$, whose indices pairwise differ by at least 3.

Solution.
The recurrence is the easy half. Condition on the last step. If the last step is 1, the rest is any climb of $n-1$ steps. If the last step is 2 and $n \geq 3$, the step before is 1, and the rest is any climb of $n-3$ steps. So $c_n = c_{n-1} + c_{n-3}$ for $n \geq 3$, with $c_0 = c_1 = 1$ and $c_2 = 2$, giving $1, 2, 3, 4, 6, 9, 13, 19, \ldots$ for $k \geq 1$, increasing.
We prove by strong induction on $N \geq 1$ that N has such a representation in which every index j satisfies $c_j \leq N$. Let $c_k \leq N < c_{k+1}$. If $N = c_k$ we are done. Otherwise $k \geq 3$, since $c_2 - c_1 = c_3 - c_2 = 1$, and

$$0 < N - c_k < c_{k+1} - c_k = c_{k-2}.$$

By the induction hypothesis $N - c_k$ is a sum of terms c_j with $c_j < c_{k-2}$, so $j \leq k-3$. Adding c_k keeps the indices at least 3 apart. ■
Remark. With gap 2 and Fibonacci numbers this is Zeckendorf's theorem. The greedy choice works because the recurrence says exactly how much room is left.

Common traps.

- A repeated root r needs nr^n as well as r^n, or two initial values cannot be matched.
- Lemma 2 divides by a_n, so prove first that no term is 0.
- A pattern in the first terms is a conjecture until you prove the recurrence behind the pattern.
- Eventual periodicity is not periodicity. Check that the recurrence can be run backwards.
- In a bound, name each inequality you use, such as $H_N \leq 1 + \ln N$.

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2 Practice problems

The problems are roughly in order of difficulty, from ⋆⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆⋆ Let $a_1 = a_2 = 1$ and $a_{n+2} = \frac{a_{n+1}^2 + a_{n+1} + 1}{a_n}$ for $n \geq 1$. Prove that every term is an integer, and that $3 \mid a_n$ if and only if $3 \mid n$.

Problem 2 ⋆⋆⋆ Find all real numbers x_0 for which the sequence defined by $x_{n+1} = x_n^2 - 2$ has infinitely many negative terms.

Problem 3 ⋆⋆⋆ Find all real numbers c for which there is a bounded sequence of integers $a_0, a_1, a_2, \ldots$, not all zero, with $a_{n+2} = c\,a_{n+1} - a_n$ for all $n \geq 0$.

Problem 4 ⋆⋆⋆⋆ Let a_1 be a positive integer and $a_{n+1} = a_n + \left\lfloor \frac{a_n}{\lfloor \sqrt{a_n} \rfloor} \right\rfloor$ for $n \geq 1$. Determine all a_1 for which the sequence contains a perfect square, and prove that the sequence then contains infinitely many.

Problem 5 ⋆⋆⋆⋆ Let α be the largest real root of $x^3 - 7x^2 + 1 = 0$. Prove that $\lfloor \alpha^n \rfloor$ is not a perfect square for any positive integer n.

Problem 6 ⋆⋆⋆⋆ Let $s_1 = 1$, $s_2 = 1$, $s_3 = 4$ and $s_n = s_{n-1} + s_{n-3}$ for $n \geq 4$. Prove that $s_p - 1$ is divisible by p for every prime p.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **AIME 1985 Problem 5** ⋆⋆
   Write the first few terms in terms of the first two.
2. **AIME II 2005 Problem 11** ⋆⋆
   See how the product of two consecutive terms changes from one step to the next.
3. **BMO2 2014/15 Problem 1** ⋆⋆
   A trigonometric substitution turns the recurrence into an addition of angles.
4. **Serbia 1998 (PEN, recursive sequences)** ⋆⋆
   Square the recurrence, and compare two consecutive steps to get a linear recurrence.
5. **AIME I 2009 Problem 13** ⋆⋆⋆
   Subtract consecutive instances of the recurrence and look at what must vanish.
6. **Baltic Way 2005 Problem 9** ⋆⋆⋆
   Set up a recurrence for full tilings together with tilings missing a corner, then work modulo 3.
7. **Baltic Way 2005 Problem 3** ⋆⋆⋆
   Bound each term of the sum by a difference that telescopes.
8. **Baltic Way 2005 Problem 17** ⋆⋆⋆
   A linear substitution turns the recurrence into a product of earlier terms.

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9. **IMO 2014 Problem 1** ⋆⋆⋆

   Define a sequence from the partial sums whose sign decides the inequality, and show that the sequence decreases at every step.

10. **IMO 2010 Problem 6** ⋆⋆⋆⋆

    Compare a_n with n times the largest ratio a_l/l, and show that the differences take finitely many values.

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4 Hints

1. The quantity $a_{n+2}a_n - a_{n+1}^2 - a_{n+1}$ is constant. Subtract two consecutive instances, as in Lemma 2.

2. If $|x_0| \le 2$, write $x_0 = 2\cos 2\pi t$ and look at the binary digits of t.

3. First show that c is rational. If $c = \frac{p}{q}$ in lowest terms, which terms must be divisible by q? Then bound $|c|$ with the characteristic roots.

4. Write $a_n = m^2 + r$ with $m = \lfloor \sqrt{a_n} \rfloor$ and $0 \le r \le 2m$. Find what two steps do to r when $r < m$.

5. Show that $\lfloor \alpha^n \rfloor = s_n - 1$, where s_n is the sum of the nth powers of the three roots. Then use one modulus for even n and another for odd n.

6. For $n \ge 4$, s_n counts the tilings of a ring of n cells by single cells and blocks of three consecutive cells. Rotate the ring by one cell.

5 Solutions

Problem 1

Let $a_1 = a_2 = 1$ and $a_{n+2} = \frac{a_{n+1}^2 + a_{n+1} + 1}{a_n}$ for $n \ge 1$. Prove that every term is an integer, and that $3 \mid a_n$ if and only if $3 \mid n$.

This is Lemma 2 with an extra term, and the only question is whether the subtraction still works. By induction every term is a positive real number. For $n \ge 1$ the definition reads

$$a_{n+2}a_n - a_{n+1}^2 - a_{n+1} = 1.$$

Equating this with the same relation for $n+1$ gives $a_{n+3}a_{n+1} + a_{n+1}^2 + a_{n+1} = a_{n+2}^2 + a_{n+2}a_n + a_{n+2}$, that is

$$a_{n+1}(a_{n+3} + a_{n+1} + 1) = a_{n+2}(a_{n+2} + a_n + 1).$$

So $k_n = \frac{a_{n+2} + a_n + 1}{a_{n+1}}$ satisfies $k_{n+1} = k_n$. As $a_3 = 3$, $k_1 = \frac{3+1+1}{1} = 5$, and

$$a_{n+2} = 5a_{n+1} - a_n - 1 \quad (n \ge 1).$$

Since a_1 and a_2 are integers, induction shows that every term is an integer. The sequence begins $1, 1, 3, 13, 61, 291$.

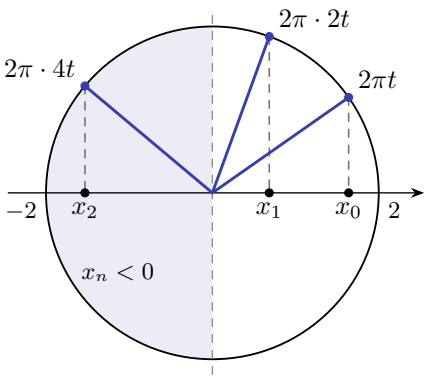
Modulo 3 the recurrence reads $a_{n+2} \equiv 2a_{n+1} + 2a_n + 2$, and each pair of consecutive residues fixes the next. The residues of $a_1, \dots, a_5$ are $1, 1, 0, 1, 1$, so $(a_4, a_5) \equiv (a_1, a_2)$ and the residues repeat with period 3. Hence $3 \mid a_n$ exactly when $3 \mid n$. ■

Remark. The term a_{n+1} in the numerator survives only as the constant -1. The subtraction of Lemma 2 needs nothing more than a relation of the same shape at consecutive indices.

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Problem 2

Find all real numbers x_0 for which the sequence defined by $x_{n+1} = x_n^2 - 2$ has infinitely many negative terms.



The case $|x_0| > 2$ is routine. If $|x_0| > 2$, then $x_1 = x_0^2 - 2 > 2$, and $x > 2$ implies $x^2 - 2 > 2$. So every term after x_0 is positive.

Let $|x_0| \le 2$ and write $x_0 = 2\cos 2\pi t$ with $0 \le t \le \frac{1}{2}$. Since $(2\cos u)^2 - 2 = 2\cos 2u$, induction gives $x_n = 2\cos(2\pi \cdot 2^n t)$. Let f_n be the fractional part of $2^n t$. Then $x_n < 0$ if and only if $\frac{1}{4} < f_n < \frac{3}{4}$.

If $t = \frac{p}{2^k}$ with integers p and $k \ge 0$, then $f_n = 0$ and $x_n = 2$ for all $n \ge k$, so only finitely many terms are negative.

Otherwise the binary expansion $t = 0.d_1 d_2 d_3 \ldots$ is unique and the digits are not eventually constant. Now $f_n = 0.d_{n+1}d_{n+2}\ldots$ in binary, so $f_n \in [\frac{1}{4}, \frac{1}{2}]$ when $d_{n+1}d_{n+2} = 01$ and $f_n \in [\frac{1}{2}, \frac{3}{4}]$ when $d_{n+1}d_{n+2} = 10$. The endpoints $\frac{1}{4}$ and $\frac{3}{4}$ cannot occur, as f_n is not of the form $\frac{p}{2^k}$ either. Two adjacent digits differ for infinitely many n, and each such n gives $x_n < 0$.

So the answer is $x_0 = 2\cos 2\pi t$ with t not of the form $\frac{p}{2^k}$, that is $x_0 = 2\cos r\pi$ with $r = 2t$ not of this form. ■

Remark. The substitution turns $x \mapsto x^2 - 2$ into the shift of the binary digits of t. A negative term is a pair of adjacent digits that differ, and only the angles whose digits end in zeros escape.

Answer: $x_0 = 2\cos r\pi$, where r is not of the form $\frac{p}{2^k}$ with p and k integers

Problem 3

Find all real numbers c for which there is a bounded sequence of integers $a_0, a_1, a_2, \ldots$, not all zero, with $a_{n+2} = c\,a_{n+1} - a_n$ for all $n \ge 0$.

Step 1. The recurrence can be run backwards, $a_n = c\,a_{n+1} - a_{n+2}$. So if two consecutive terms are 0, every term is 0. In particular some $a_{n+1} \neq 0$, and $c = \frac{a_{n+2}+a_n}{a_{n+1}}$ is rational.

Step 2. This is the step that needs an idea. Write $c = \frac{p}{q}$ with $q \ge 1$ and $\gcd(p,q) = 1$. From $q(a_{n+2} + a_n) = p\,a_{n+1}$ we get $q \mid a_{n+1}$ for every $n \ge 0$. So $b_n = \frac{a_{n+1}}{q}$ is again a bounded integer sequence satisfying the recurrence, and not all zero by Step 1. Repeating, $q^j \mid a_n$ for all $n \ge j$. If $q \ge 2$, choose j with q^j larger than every $|a_n|$. Then $a_n = 0$ for all $n \ge j$, which contradicts Step 1. So $q = 1$ and c is an integer.

Step 3. Suppose $|c| \ge 3$. The roots of $x^2 - cx + 1$ are real, r and $\frac{1}{r}$ with $|r| > 1$, so by Theorem 1, $a_n = Ar^n + Br^{-n}$. Boundedness forces $A = 0$, because $|A||r|^n \le |a_n| + |B|$. Then $|a_n| = |B||r|^{-n} <$

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1 for large n, so $a_n = 0$ for large n, against Step 1. Hence $|c| \le 2$.

Step 4. Each remaining value works. For $c = 2$ take $1, 1, 1, \ldots$, for $c = -2$ take $1, -1, 1, -1, \ldots$, for $c = 1$ take $0, 1, 1, 0, -1, -1, \ldots$ (period 6), for $c = 0$ take $0, 1, 0, -1, \ldots$ (period 4), and for $c = -1$ take $0, 1, -1, 0, 1, -1, \ldots$ (period 3). So $c \in \{-2, -1, 0, 1, 2\}$. ■

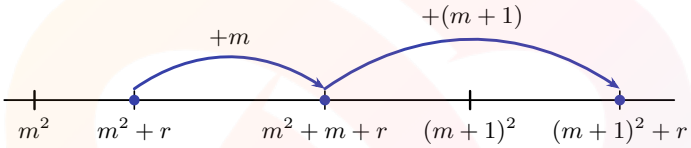
Remark. Step 2 replaces Niven's theorem, which says that if θ is a rational multiple of π and $2\cos\theta$ is rational, then $2\cos\theta \in \{-2, -1, 0, 1, 2\}$. Divisibility alone forces c to be an integer.

Answer: $c \in \{-2, -1, 0, 1, 2\}$

Problem 4

Let a_1 be a positive integer and $a_{n+1} = a_n + \left\lfloor \frac{a_n}{\lfloor \sqrt{a_n} \rfloor} \right\rfloor$ for $n \ge 1$. Determine all a_1 for which the sequence contains a perfect square, and prove that the sequence then contains infinitely many.

Every positive integer a is uniquely $m^2 + r$ with $m = \lfloor \sqrt{a} \rfloor \ge 1$ and $0 \le r \le 2m$, and then $\lfloor a/m \rfloor = m + \lfloor r/m \rfloor$.



Two steps when $r < m$. The figure shows the two steps. Then $\lfloor a/m \rfloor = m$, and the next term is $m^2 + m + r$. This is less than $(m+1)^2$, so the floor of the square root is still m, and dividing by m gives $m + 1 + \frac{r}{m}$, with floor $m + 1$. So the term after is $(m+1)^2 + r$. Hence the terms alternate between $M^2 + e$ and $M^2 + M + e$ for consecutive M, with $e = r$ fixed and $e < M$ throughout.

The other cases. If $m \le r < 2m$, then $a = m^2 + m + e$ with $e = r - m < m$, which is the second kind of term, and the step gives $(m+1)^2 + e$ as before. If $r = 2m$, then $\lfloor a/m \rfloor = m + 2$ and the next term is $(m+1)^2 + (m+1)$, the second kind of term with $e = 0$. So from a_1 or a_2 on the sequence follows the pattern, where $e = r$ if $r < m$, $e = r - m$ if $m \le r < 2m$, and $e = 0$ if $r = 2m$, for $a_1 = m^2 + r$.

Squares. The numbers $M^2 + e$ and $M^2 + M + e$ with $0 \le e < M$ lie in $[M^2, (M+1)^2)$, so such a term is a square only when the term is M^2, that is $e = 0$ and the term is of the first kind. If $e = 0$, every other term is a square. If $e \ge 1$, then a_1 is already in the pattern and no term is a square.

So the sequence contains a square if and only if $e = 0$, that is $r \in \{0, m, 2m\}$ and $a_1 \in \{m^2, m^2 + m, m^2 + 2m\}$, and then the sequence contains infinitely many squares. ■

Remark. Finding the invariant offset e is the whole problem. A good first experiment is $a_1 = 5$, which gives $5, 7, 10, 13, 17, 21, 26, \ldots$ with $e = 1$.

Answer: $a_1 = k^2$, $k^2 + k$ or $k^2 + 2k$ for a positive integer k

Problem 5

Let α be the largest real root of $x^3 - 7x^2 + 1 = 0$. Prove that $\lfloor \alpha^n \rfloor$ is not a perfect square for any positive integer n.

Two facts are standard, that the other two roots are small and that the power sums are integers. The real problem is choosing the moduli. Let $f(x) = x^3 - 7x^2 + 1$. Then $f(-\frac{1}{2}) = -\frac{7}{8}$, $f(0) = 1$, $f(\frac{1}{2}) = -\frac{5}{8}$, $f(6) = -35$ and $f(7) = 1$, so the roots satisfy

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$$-\tfrac{1}{2} < \gamma < 0 < \beta < \tfrac{1}{2}, \qquad 6 < \alpha < 7.$$

Also $\beta + \gamma = 7 - \alpha > 0$, so $\beta > |\gamma|$. For $n \geq 1$ this gives $\beta^n + \gamma^n \geq \beta^n - |\gamma|^n > 0$ and $\beta^n + \gamma^n < 2 \cdot 2^{-n} \leq 1$.

Let $s_n = \alpha^n + \beta^n + \gamma^n$. Each root satisfies $x^{n+3} = 7x^{n+2} - x^n$, so $s_{n+3} = 7s_{n+2} - s_n$, with $s_0 = 3$, $s_1 = 7$ and $s_2 = 7^2 - 2 \cdot 0 = 49$ by Vieta. So every s_n is an integer, $s_n - 1 < \alpha^n < s_n$, and $\lfloor \alpha^n \rfloor = s_n - 1$.

Even n. Modulo 5 the recurrence is $s_{n+3} \equiv 2s_{n+2} - s_n$. The residues of $s_0, s_1, \ldots, s_6$ are $3, 2, 4, 0, 3, 2, 4$, so the residues have period 4, and $s_n - 1$ is 2 or 3 modulo 5 for $n \equiv 0$ or $2 \pmod 4$. A square is 0, 1 or 4 modulo 5.

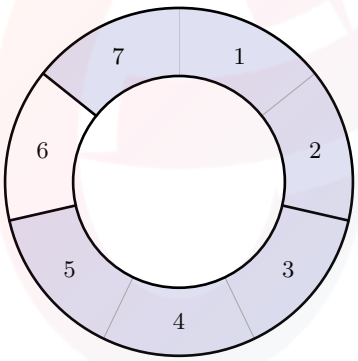
Odd n. Modulo 7 the recurrence is $s_{n+3} \equiv -s_n$, so the residues of $s_0, s_1, s_2, \ldots$ run $3, 0, 0, 4, 0, 0, 3, \ldots$ with period 6. For odd n this gives $s_n - 1 \equiv 6$ when $3 \nmid n$ and $s_n - 1 \equiv 3$ when $n \equiv 3 \pmod 6$. A square is 0, 1, 2 or 4 modulo 7.

So $\lfloor \alpha^n \rfloor = s_n - 1$ is never a square. ■

Remark. Neither modulus works alone. Modulo 5 the odd n give the residues 1 and 4, and modulo 7 the index $n = 6$ gives 2, all squares.

Problem 6

Let $s_1 = 1$, $s_2 = 1$, $s_3 = 4$ and $s_n = s_{n-1} + s_{n-3}$ for $n \geq 4$. Prove that $s_p - 1$ is divisible by p for every prime p.



The recurrence alone gives no reason for p to divide $s_p - 1$. The idea is to find something that s_n counts and then rotate. *A model for s_n.* Place $n \geq 4$ cells around a circle and tile them by single cells and by blocks of three consecutive cells. Let c_n be the number of tilings, and let t_m be the number of tilings of a row of m cells by the same pieces, so $t_0 = t_1 = t_2 = 1$ and $t_m = t_{m-1} + t_{m-3}$ for $m \geq 3$ (condition on the last piece). In a ring tiling, cell 1 is either a single cell, leaving a row of $n - 1$ cells, or lies in one of the three blocks $\{n-1, n, 1\}$, $\{n, 1, 2\}$, $\{1, 2, 3\}$, which are different as $n \geq 4$, each leaving a row of $n - 3$ cells. So

$$c_n = t_{n-1} + 3t_{n-3}.$$

For $n \geq 7$ the recurrence for t gives

$$c_n - c_{n-1} - c_{n-3} = (t_{n-1} - t_{n-2} - t_{n-4}) + 3(t_{n-3} - t_{n-4} - t_{n-6}) = 0.$$

Also $s_4 = 5$, $s_5 = 6$, $s_6 = 10$, while $c_4 = t_3 + 3t_1 = 5$, $c_5 = t_4 + 3t_2 = 6$ and $c_6 = t_5 + 3t_3 = 10$. Hence $c_n = s_n$ for all $n \geq 4$.

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Small primes. $s_2 - 1 = 0$ and $s_3 - 1 = 3$.

Rotation. Let $p \geq 5$ be prime and let ρ rotate the ring by one cell. Then ρ permutes the c_p tilings and ρ^p is the identity, so each orbit $\{T, \rho T, \rho^2 T, \dots\}$ has size dividing p, that is 1 or p. If $\rho T = T$, then T is fixed by every rotation. Should T contain a block $\{j, j+1, j+2\}$, the block $\{j+1, j+2, j+3\}$ would also be in T, and the two blocks are different but overlap, which is impossible. So the only fixed tiling is the one by single cells, and

$$s_p = c_p = 1 + p \cdot (\text{number of orbits of size } p) \equiv 1 \pmod{p}.$$

So p divides $s_p - 1$. ■

Remark. In algebraic terms $s_n = \alpha^n + \beta^n + \gamma^n$ for the roots of $x^3 - x^2 - 1$, and $s_p \equiv (\alpha + \beta + \gamma)^p = 1$. The rotation proof needs no algebra, only a model in which the recurrence counts something.

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